2025 – 2026 Semester II
COMP2432 Operating Systems
The Hong Kong Polytechnic University

Programming Assignment

Note: *All assignment queries should be posted on BlackBoard Discussion Forum*

Additionally, you are also welcome to discuss the assignment with the following TA's

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Posted on Saturday, January 24, 2026

Objective: The objective of this programming assignment is to practice and apply knowledge of BASH programming as a script language to process on a collection of files to extract and consolidate information.

Late submission penalty

1. 10% is deducted for each day that the work is late. The penalty will be applied up to a maximum number of three days after and including the submission deadline day. After three days the work will be marked at zero.
2. **This assignment is individual work. All work must be done on your own. Plagiarism is serious offence. Copying code from web resources is prohibited as well. Any plagiarism case (for both the copier and the copiee) will be given a ZERO mark in this assignment.**

---------- Submission Requirement ----------

1. Submit your program via BlackBoard.
2. Since the BlackBoard submission system disallows the submission of a script file (with extension **`.sh`**) for security reason, please submit your work under the name SID.txt in which SID is your student ID. For example, if your student ID is 24681379D, name the program as **`24681379D.txt`**.
3. Note that a line in the Linux system is terminated by a character of <LF> only, but a line in the Windows system is terminated by a pair of characters <CR><LF>. Make sure that you have removed the extra <CR> character if you use a PC to create the file, or else the script program cannot be executed on **`apollo`** or **`apollo2`**.
4. **Warning: Any runtime error will lead to 50% of deduction of points and partial points will be awarded depending upon the logic of what you have coded. Therefore, if you are unable to complete the whole program, try to accomplish part of the tasks and make sure that it can be run.**
5. **Any wrong file naming and submission will be given a ZERO mark in this assignment.**
6. The deadline for this assignment is **23:59:00 Sunday, February 15, 2026**.

Problem: Analyzing Player's Performance for a Basketball Team

Objective: The objective of this assignment is to search through a list of game play data files, consolidate and generate the performance of some specific player for further analysis.

Scenario: You are given a set of game play data files, each is of the format described below. Each file is retrieved by an internet bot program about the summary information of a game from the official NBA web site. You are to write a **`bash`** script program to process the summary files in order to produce useful statistics about a given player. This will help in understanding player performance and identifying players for potential trade consideration in order to enhance the competitiveness of the team.

Here are sample files about the NBA games: Los Angeles Lakers (LAL) at Denver Nuggets (DEN) on Jan 20, San Antonio Spurs (SAS) at Houston Rockets on Jan 20, and Los Angeles Lakers (LAL) at San Antonio Spurs (SAS) on Jan 7. The first line of the file indicates the visiting team. The second line contains the names of the key fields. The subsequent lines contain information about individual players and their performance. DNP means that the player **D**id **N**ot **P**lay in the game. The final line summarizes the team performance. The second half of the file contains similar information about the home team. You can assume that all the text files do not contain any error. Furthermore, all non-English characters have been converted into English characters.

nba260120LALDEN.txt

```
Los Angeles Lakers
PLAYER MIN    FGM    FGA    FG%    3PM    3PA    3P%    FTM    FTA    FT%    OREB   DREB   REB    AST
       STL    BLK    TO     PF     PTS    +/-
LeBron James F 34:25        8      15     53.3   1      1      100    2      6      33.3   1      8
       9      8      0      0      6      2      19     13
Jake LaRavia F 26:45        3      5      60.0   0      2      0.0    1      1      100    0      1
       1      2      0      0      3      2      7      4
Deandre Ayton C 14:49       2      5      40.0   0      0      0.0    0      0      0.0    1      7
       8      0      0      0      0      1      4      -1
Marcus Smart G 31:15        4      9      44.4   3      6      50.0   4      5      80.0   1      1
       2      1      0      0      0      2      15     6
Luka Doncic G 35:39 12      21     57.1   3      9      33.3   11     12     91.7   0      13     13
       10     2      0      3      2      38     -2
Rui Hachimura 28:15 4       13     30.8   1      3      33.3   0      0      0.0    3      3      6
       1      1      0      0      2      9      -2
Jaxson Hayes 22:52  4       5      80.0   0      0      0.0    1      4      25.0   1      4      5
       0      0      2      0      2      9      16
Gabe Vincent 17:28  1       2      50.0   1      2      50.0   2      3      66.7   0      0      0
       0      0      0      0      3      5      3
Jarred Vanderbilt 17:47     0      2      0.0    0      1      0.0    0      0      0.0    1      3
       4      2      0      0      0      1      0      -2
Drew Timme 10:45    4       6      66.7   1      1      100    0      1      0.0    2      1      3
       0      0      0      0      1      9      5
Kobe Bufkin DNP - Coach's Decision
Bronny James DNP - Coach's Decision
Maxi Kleber DNP - Coach's Decision
Dalton Knecht DNP - Coach's Decision
TOTALS 42      83     50.6   10     25     40.0   21     32     65.6   10     41     51     24     3
       2      12     18     115    8
Denver Nuggets
PLAYER MIN    FGM    FGA    FG%    3PM    3PA    3P%    FTM    FTA    FT%    OREB   DREB   REB    AST
       STL    BLK    TO     PF     PTS    +/-
Spencer Jones F 36:47       6      9      66.7   4      7      57.1   0      0      0.0    0      5
       5      0      1      1      1      5      16     6
Peyton Watson F 37:45       7      17     41.2   4      8      50.0   0      0      0.0    1      4
       5      3      3      1      3      3      18     -6
Aaron Gordon C 31:51        6      9      66.7   2      4      50.0   4      8      50.0   1      3
       4      5      0      0      0      1      18     -8
Jalen Pickett G 26:14       3      9      33.3   1      2      50.0   0      0      0.0    1      5
       6      4      1      0      0      2      7      -4
Jamal Murray G 36:05        10     21     47.6   4      8      50.0   4      4      100    0      2
```

```
        2       11      0       0       1       2       28      -3
Tim Hardaway Jr.  30:43     2       12      16.7    0       8       0.0     6       7       85.7    0       2
        2       2       1       3       0       3       10      0
Zeke Nnaji 18:35    3       4       75.0    0       0       0.0     0       0       0.0     1       2       3
        0       0       2       0       2       6       -4
Bruce Brown 22:00   2       6       33.3    0       0       0.0     0       0       0.0     1       3       4
        2       0       0       0       3       4       -21
DaRon Holmes II DNP - Coach's Decision
Julian Strawther DNP - Coach's Decision
Hunter Tyson DNP - Coach's Decision
TOTALS 39       87      44.8    15      37      40.5    14      19      73.7    5       26      31      27      6
        7       5       21      107     -8
```

nba260120SASHOU.txt

```
San Antonio Spurs
PLAYER MIN      FGM     FGA     FG%     3PM     3PA     3P%     FTM     FTA     FT%     OREB    DREB    REB     AST
        STL     BLK     TO      PF      PTS     +/-
Julian Champagnie F 34:10   8       17      47.1    8       16      50.0    3       3       100     1       7
        8       2       2       0       0       3       27      5
Harrison Barnes F 24:37     2       6       33.3    2       5       40.0    0       0       0.0     0       2
        2       1       1       1       1       0       6       -2
Victor Wembanyama C 29:42   5       21      23.8    0       7       0.0     4       4       100     5       5
        10      2       0       0       1       4       14      -14
De'Aaron Fox G 32:12        6       15      40.0    2       6       33.3    0       0       0.0     0       3
        3       5       1       0       3       2       14      -7
Stephon Castle G 34:53      5       19      26.3    2       8       25.0    1       1       100     4       3
        7       8       1       1       3       2       13      -12
Luke Kornet 23:53   4       5       80.0    0       0       0.0     0       0       0.0     2       4       6
        1       0       2       0       3       8       3
Dylan Harper 22:37  4       9       44.4    0       3       0.0     0       0       0.0     0       3       3
        5       0       1       1       0       8       2
Keldon Johnson 27:12        5       9       55.6    0       3       0.0     2       2       100     2       4
        6       0       2       0       0       1       12      -2
Carter Bryant 10:44 2       4       50.0    0       2       0.0     0       0       0.0     0       0       0
        0       0       0       0       1       4       2
Bismack Biyombo DNP - Coach's Decision
Jordan McLaughlin DNP - Coach's Decision
Kelly Olynyk DNP - Coach's Decision
Jeremy Sochan DNP - Coach's Decision
Lindy Waters III DNP - Coach's Decision
TOTALS 41       105     39.0    14      50      28.0    10      10      100     14      31      45      24      7
        5       9       16      106     -5
Houston Rockets
PLAYER MIN      FGM     FGA     FG%     3PM     3PA     3P%     FTM     FTA     FT%     OREB    DREB    REB     AST
        STL     BLK     TO      PF      PTS     +/-
Kevin Durant F 39:21        7       12      58.3    2       4       50.0    2       2       100     0       4
        4       7       1       1       4       2       18      1
Jabari Smith Jr.  F 35:10   7       13      53.8    2       5       40.0    1       3       33.3    1       3
        4       1       0       1       0       2       17      1
Alperen Sengun C 33:34      7       13      53.8    1       3       33.3    5       5       100     0       13
        13      9       0       2       4       0       20      -2
Josh Okogie G 16:37 2       2       100     1       1       100     0       3       0.0     1       0       1
        0       1       0       0       2       5       -9
Amen Thompson G 39:14       6       14      42.9    0       2       0.0     4       5       80.0    0       10
        10      6       1       2       3       0       16      7
Tari Eason 19:49    3       6       50.0    0       0       0.0     0       0       0.0     2       4       6
        1       1       1       3       2       6       -3
Dorian Finney-Smith 14:53   1       3       33.3    0       2       0.0     0       0       0.0     1       0
        1       1       0       0       0       2       2       9
Clint Capela 12:13  2       3       66.7    0       0       0.0     2       2       100     1       1       2
        0       0       1       0       1       6       6
Reed Sheppard 29:10 8       17      47.1    4       10      40.0    1       2       50.0    0       2       2
        4       1       1       0       0       21      15
JD Davison DNP - Coach's Decision
Jeff Green DNP - Coach's Decision
Aaron Holiday DNP - Coach's Decision
Jae'Sean Tate DNP - Coach's Decision
TOTALS 43       83      51.8    10      27      37.0    15      22      68.2    6       37      43      29      5
        9       14      11      111     5
```

Los Angeles Lakers

PLAYER	MIN	FGM	FGA	FG%	3PM	3PA	3P%	FTM	FTA	FT%	OREB	DREB	REB	AST
	STL	BLK	TO	PF	PTS	+/-								
Jarred Vanderbilt F	18:51	2	6	33.3	0	3	0.0	0	0	0.0	0	0		
	0	0	1	0	1	3	4	0						
Jake LaRavia F	38:35	6	13	46.2	4	9	44.4	0	0	0.0	4	3		
	7	1	1	0	1	2	16	-13						
Deandre Ayton C	28:54	4	9	44.4	0	0	0.0	1	2	50.0	1	5		
	6	1	0	1	1	2	9	-9						
Marcus Smart G	34:29	1	6	16.7	0	3	0.0	0	0	0.0	0	7		
	7	1	3	0	4	3	2	-12						
Luka Doncic G	38:20	13	26	50.0	2	9	22.2	10	16	62.5	0	10	10	
	10	3	0	7	3	38	-7							
Dalton Knecht	19:10	1	6	16.7	0	5	0.0	0	0	0.0	0	3	3	
	0	0	1	3	1	2	-15							
Jaxson Hayes	18:15	4	4	100	0	0	0.0	2	2	100	2	5	7	
	0	0	0	0	3	10	-7							
Gabe Vincent	18:08	2	8	25.0	2	7	28.6	0	0	0.0	1	1	2	
	2	0	0	0	2	6	-10							
Maxi Kleber	12:19	0	3	0.0	0	0	0.0	1	2	50.0	2	1	3	
	0	1	0	0	2	1	-3							
Nick Smith Jr.	10:26	1	4	25.0	1	2	50.0	0	0	0.0	0	0		
	0	1	0	0	0	0	3	-4						
Chris Manon	0:51	0	0	0.0	0	0	0.0	0	0	0.0	0	0	0	
	0	0	0	0	0	0	0							
Bronny James	0:51	0	0	0.0	0	0	0.0	0	0	0.0	0	0	0	
	0	0	0	0	0	0	0							
Drew Timme	0:51	0	1	0.0	0	1	0.0	0	0	0.0	0	0	0	
	0	0	0	0	0	0	0							
TOTALS	34	86	39.5	9	39	23.1	14	22	63.6	10	35	45	16	9
	2	17	21	91	-16									

San Antonio Spurs

PLAYER	MIN	FGM	FGA	FG%	3PM	3PA	3P%	FTM	FTA	FT%	OREB	DREB	REB	AST
	STL	BLK	TO	PF	PTS	+/-								
Julian Champagnie F	33:34	3	11	27.3	0	6	0.0	5	6	83.3	3	6		
	9	3	0	0	2	1	11	18						
Harrison Barnes F	29:53	2	5	40.0	0	1	0.0	3	4	75.0	0	0		
	0	6	1	0	0	2	7	16						
Luke Kornet C	21:11	2	5	40.0	0	0	0.0	3	4	75.0	3	3	6	
	1	2	0	1	2	7	7							
De'Aaron Fox G	34:12	5	14	35.7	0	6	0.0	4	4	100	0	3		
	3	5	1	0	3	2	14	12						
Stephon Castle G	29:07	6	10	60.0	0	2	0.0	3	3	100	0	5		
	5	4	2	1	2	5	15	9						
Dylan Harper	19:57	1	6	16.7	0	2	0.0	0	0	0.0	0	4	4	
	2	2	0	0	1	2	-2							
Keldon Johnson	27:37	11	13	84.6	2	3	66.7	3	3	100	1	5		
	6	0	0	0	1	1	27	14						
Victor Wembanyama	25:58	7	15	46.7	0	3	0.0	2	4	50.0	1	13		
	14	2	2	4	4	3	16	9						
Jeremy Sochan	11:28	2	2	100	2	2	100	0	0	0.0	0	0	0	
	1	0	0	0	3	6	-1							
Carter Bryant	4:30	1	1	100	0	0	0.0	0	0	0.0	0	2	2	
	0	0	0	1	0	2	-2							
Lindy Waters III	0:51	0	0	0.0	0	0	0.0	0	0	0.0	0	0		
	0	0	0	0	0	0	0	0						
Bismack Biyombo	0:51	0	0	0.0	0	0	0.0	0	0	0.0	0	0		
	0	0	0	0	0	0	0	0						
David Jones Garcia	0:51	0	0	0.0	0	0	0.0	0	0	0.0	0	1		
	1	0	0	0	0	0	0	0						
Jordan McLaughlin DNP - Coach's Decision														
Kelly Olynyk DNP - Coach's Decision														
TOTALS	40	82	48.8	4	25	16.0	23	28	82.1	8	42	50	24	10
	5	14	20	107	16									

Explanation of the fields:

PLAYER	Name of player (starter as **Guard** / **Forward** / **Center**)	**OREB**	Offensive rebounds
MIN	Number of minutes played	**DREB**	Defensive rebounds
FGM	Field goals made	**REB**	Rebounds
FGA	Field goals attempted	**AST**	Assists
FG%	Field goals percentage	**STL**	Steals
3PM	Three-point field goals made	**BLK**	Blocks
3PA	Three-point field goals attempted	**TO**	Turnovers
3P%	Three-point field goals percentage	**PF**	Personal fouls
FTM	Free throws made	**PTS**	Points
FTA	Free throws attempted	**+/-**	A form of player efficiency
FT%	Free throws percentage		

Your **bash** program (**performance.sh**) should consolidate the player performance through all the available files in the current directory, with names **nba**YYMMDD*VisitingTeamHomeTeam*.**txt**. The first argument contains the player name, and the subsequent arguments the performance statistics to be retrieved. Note that some statistics can be easily computed by aggregating them (e.g. **FGM**, **FGA**), but some others need more careful calculation (e.g. **FG%**). All statistics should be provided at one decimal place. You can assume that there is no mistake in the input line. Here are some possible scenarios in executing your program with the corresponding output results based on the only 3 sample files above. Further analysis and comparison on a group of players can then be easily made by writing additional script programs utilizing your script program as an ingredient.

```
performance.sh Doncic PTS REB AST STL
PTS 38.0
REB 11.5
AST 10.0
STL 2.5
performance.sh 'LeBron James' FGM 3PM FTM PTS
FGM 8.0
3PM 1.0
FTM 2.0
PTS 19.0
performance.sh Wembanyama PTS FG% 3P% FT%
PTS 15.0
FG% 33.3
3P% 0.0
FT% 75.0
```

Requirements:

Basic level: correct aggregation of some simple statistics.

Normal level: correct calculation of all statistics.

Advanced level: provision with explanation for some additional functionality implemented beyond just correctness, e.g. report the statistics of several players with the same search name (like James), report separate statistics of a player who has played in more than one team.

Please provide appropriate comments and check to ensure that your program can be executed properly under the Linux (**apollo** or **apollo2**) environment before submission. Your program should run on any collection of data files in correct format.

Note: There is a program called **awk**, which is an arbitrary-precision mathematical calculator utility in Unix/Linux capable of doing complex calculations rather than standard integer arithmetic as in **bash**.